

Nicolás Bodnariuk

Data Scientist — Climate Analytics

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🌐 <https://nbodnariuk.github.io>

🐙 [nbodnariuk](https://github.com/nbodnariuk)

Profile

Data scientist with a PhD in ocean and atmospheric sciences, specialised in applied analytics, uncertainty quantification, predictive modelling and geospatial data processing. Experienced working with real-world large environmental datasets, numerical modelling workflows and statistical analysis using Julia, MATLAB and Python. Comfortable operating across research and business-facing environments, including private-sector consulting.

Technical Skills

Programming Julia, MATLAB, Python

Data analysis clustering, large-data set statistical analysis, time series analysis, spatial field pattern extraction

Modelling Monte Carlo, stochastic modelling, predictive analytics, dynamical systems modelling

Professional Experience

2021–Present **Independent Consultant (Environmental Analytics)**, *Bahitek SRL*, Buenos Aires, Argentina

- Built Monte Carlo uncertainty propagation workflows integrating HYSYS and MATLAB to quantify output variability in industrial pumping networks.
- Processed meteorological data streams (Python, Julia) for pollutant dispersion modelling used by Profertil, Mega, Renova, Viterro and TGS.
- Performed statistical analysis of operational data identifying drivers of temperature-humidity anomalies impacting industrial operations (Aluar plant).

2024–Present **Postdoctoral Researcher (Data-driven modelling)**, *CNRS – LMD*, Paris, France

- Developed a fully custom Quasi-Geostrophic model in Julia to simulate ocean dynamics and analyse nonlinear regime transitions using topological methods.
- Processed multi-terabyte Earth observation datasets (netCDF/xarray) to identify climate variability patterns using EOFs and MSSA.

2016–2024 **Doctorand and postdoctoral researcher — (big data)**, *CONICET*, Buenos Aires, Argentina

- Applied statistical methods to detect change points, spatiotemporal trends and climate drivers affecting regional ocean circulation on interannual timescales.

Selected Applied Projects

Larval Transport Modeling	Developed a numerical particle-tracking model to simulate squid larval dispersal using ocean current fields. This work integrated physical oceanographic variables (temperature, salinity, currents) and environmental drivers to support fisheries stock assessment.
Topological Data Analysis	Applied topological data methods (attractor reconstruction) to extract fundamental dynamical structure and low-dimensional drivers of variability in simplified models of ocean circulation.

Education

2022 **Ph.D. Atmospheric and Oceanic Sciences**, *University of Buenos Aires*

2015 **M.Sc. Physical Oceanography**, *Universidad Nacional del Sur*

Selected Publications

Author of 15+ peer-reviewed articles in applied modelling, environmental analytics and dynamical systems. Full list available at nbodnariuk.github.io.

Languages

Spanish (native), English (C2), French (C1), German (B1)